Lecture 5

MLE and Multivariate Gaussians

Introducing Maximum Likelihood Estimation, and Gaussian Mixture Models

EECS 189/289, Spring 2026 @ UC Berkeley

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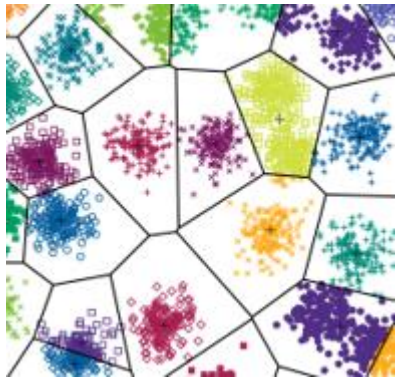
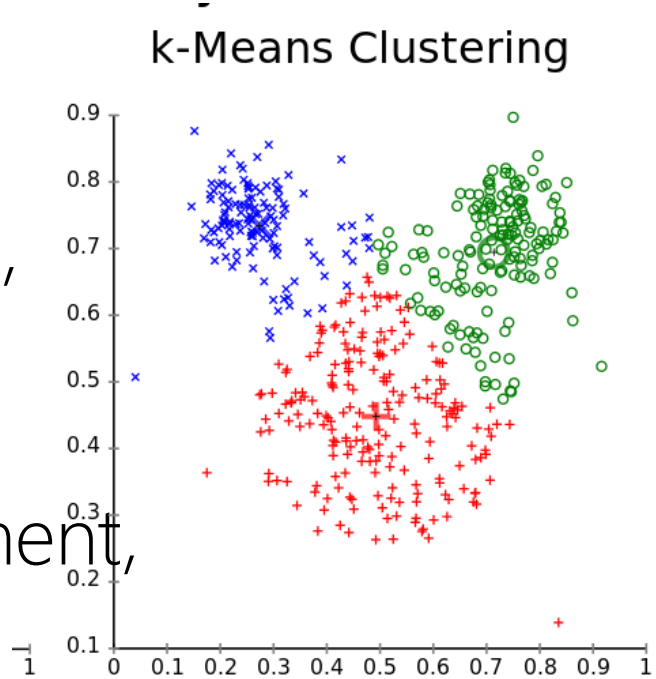
Last class: k-means clustering & weaknesses



$$\underset{S=C_1 \cup \ldots \cup C_K,\ \{c_1,\ldots,c_K\}}{\operatorname{argmin}} \sum_k \sum_{x \in C_k} \|x - c_k\|^2$$

Not probabilistically framed, so hard to understand assumptions and their implications:

- implicitly corresponds to clusters of similar volume, that are roughly isotropic (a "blobby tessellation").
- also hard to generalize
- each step in the optimization has a "hard" assignment, which feels "brittle"



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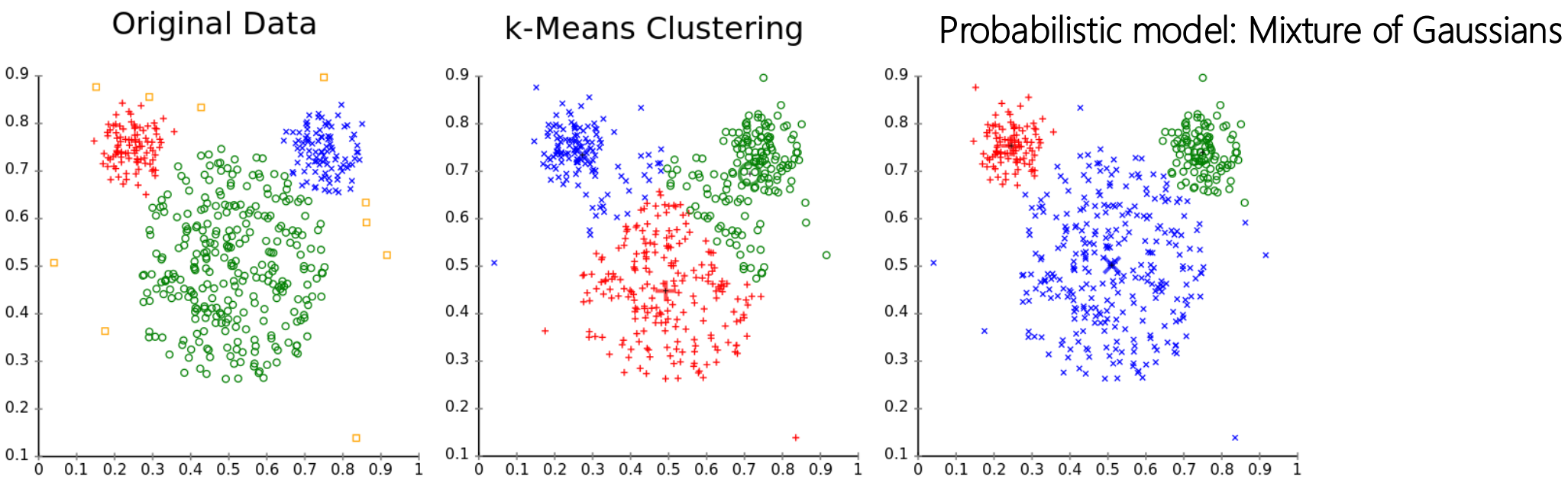
Probabilistic model of K-means that fixes these issues: *Mixture of Gaussians (Gaussian Mixture Model)*



https://en.wikipedia.org/wiki/K-means_clustering

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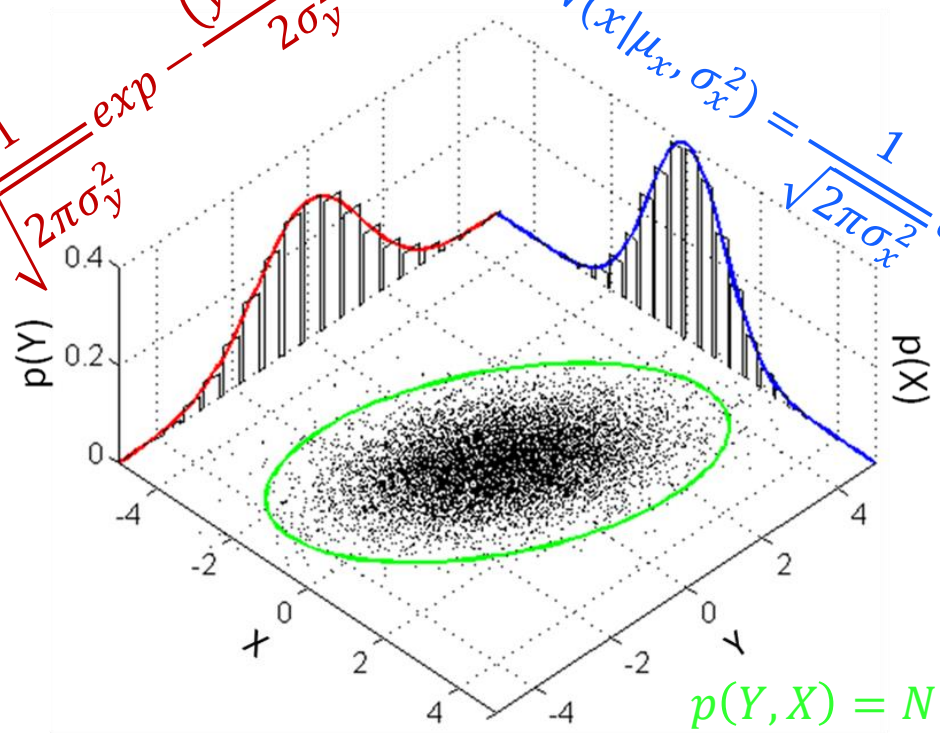
Probabilistic model of K-means that fixes these issues: *Mixture of Gaussians (Gaussian Mixture Model)*

We will need to estimate parameters of a *Mixture of Gaussians*, requiring *Multivariate Gaussians*

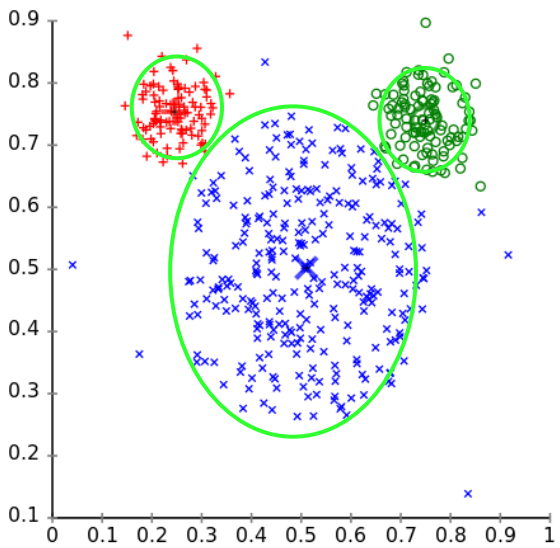
$$Y \sim N(y|\mu_y, \sigma_y^2) = \frac{1}{\sqrt{2\pi\sigma_y^2}} exp - \frac{(y-\mu_y)^2}{2\sigma_y^2}$$

$$X \sim N(x|\mu_x, \sigma_x^2) = \frac{1}{\sqrt{2\pi\sigma_x^2}} exp - \frac{(x-\mu_x)^2}{2\sigma_x^2}$$



$p(Y, X) = N(?)$ ← multivariate Gaussian

Mixture of Gaussians



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Probabilistic model of K-means that fixes these issues: *Mixture of Gaussians (Gaussian Mixture Model)*

We will need to estimate parameters of a *Mixture of Gaussians*, requiring *Multivariate Gaussians* & *Maximum Likelihood Estimation.*

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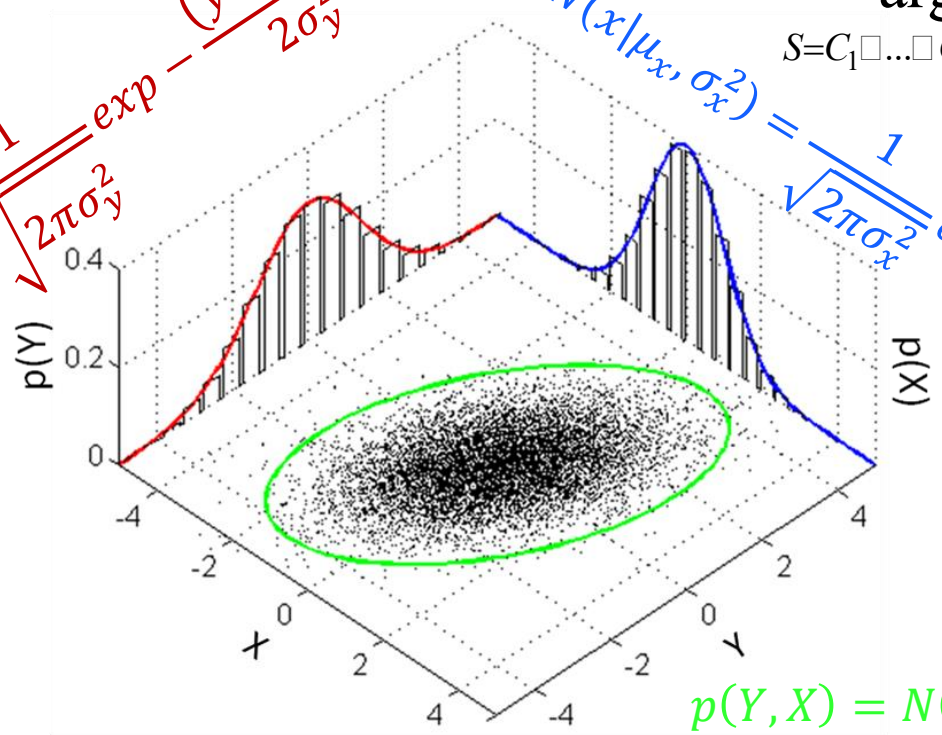
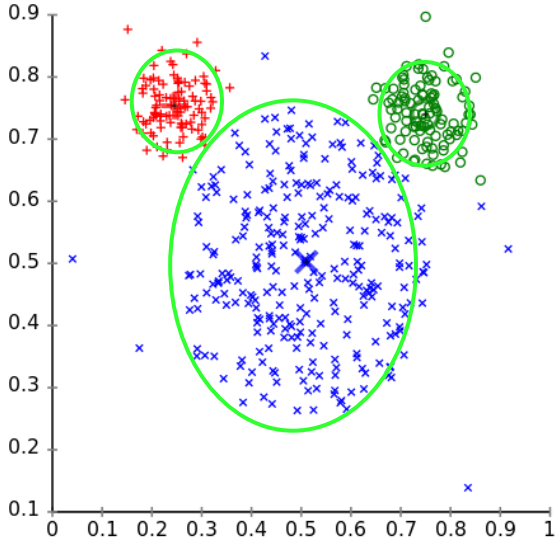
$$\cancel{\underset{S=C_1 \square \ldots \square C_K,\ \{c_1,\ldots,c_K\}}{\text{argmin}} \sum_k \sum_{x \square C_k} \|x - c_k\|^2}$$



Mixture of Gaussians



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Roadmap

- Maximum Likelihood Estimation (MLE)
- Multivariate Gaussians

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- Maximum Likelihood Estimation (MLE)
- Multivariate Gaussians

ML: main abstract ideas

- Training data set: $D = \{(x_i, y_i)\}_{i=1}^{N}$

$x_i \in R^D$

$y_i \in R$ or $y_i \in \{-1,1\}$

classification

regression

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ML: main abstract ideas

- Training data set: $D = \{(x_i, y_i)\}_{i=1}^{N}$ $\quad x_i \in R^D$, $y_i \in R$ or $y_i \in \{-1,1\}$

SUPERVISED

"label" provides supervision

$D = \{(x_i\ )\}_{i=1}^{N}$ $\quad x_i \in R^D$

UNSUPERVISED

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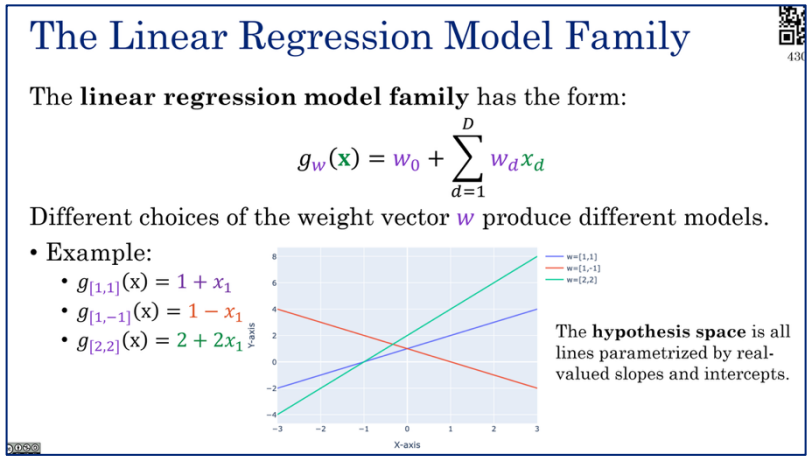
ML: main abstract ideas

- Training data set: $D = \{(x_i, y_i)\}_{i=1}^{N}$ $\quad x_i \in R^D$ $\quad y_i \in R$ or $y_i \in \{-1,1\}$

- Model class:

aka hypothesis class

$$g(x \mid w, w_0) = w^T x + w_0$$



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ML: main abstract ideas

- Training data set:

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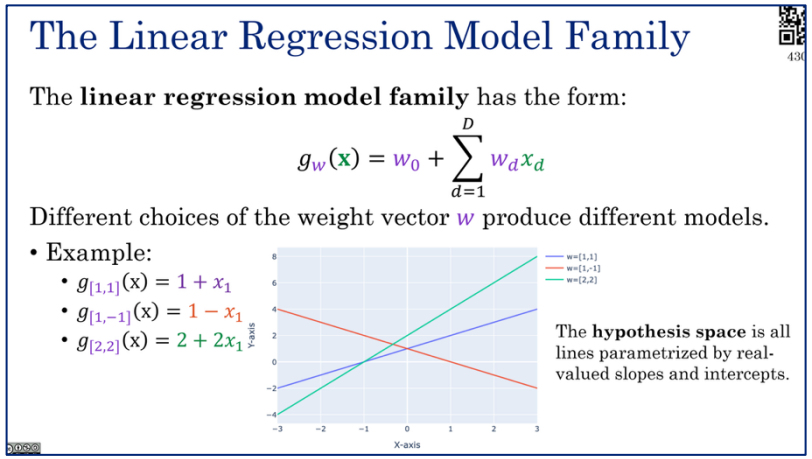

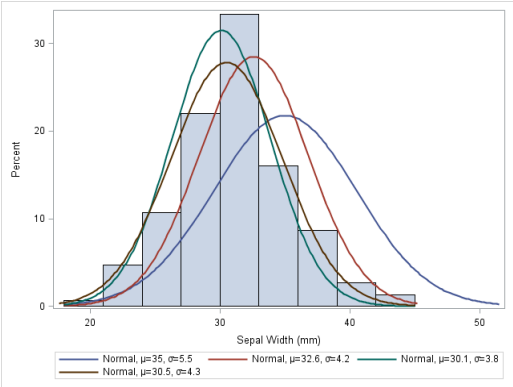

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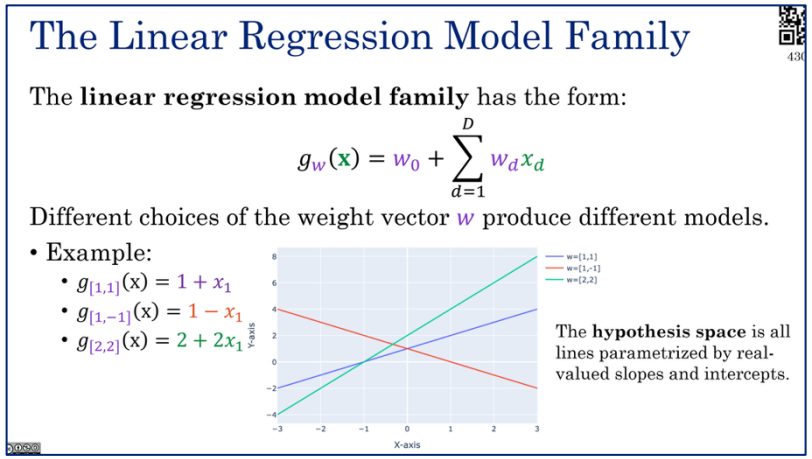
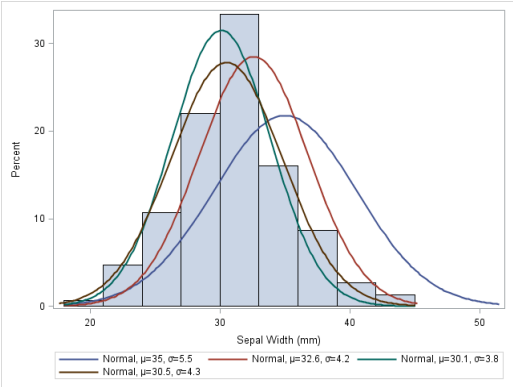


- **Optimization goal:** find "good" values of parameters (w, w_0) or $(\mu, \sigma\ )$.
- But was does "good" mean?



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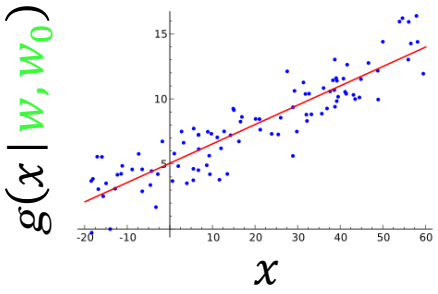

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  $g(x \mid w, w_0) = w^T x + w_0$ **Linear Regression**

  $p(x \mid \mu, \sigma) = N(x \mid \mu, \sigma)$ **Gaussian Probability Density**

- Loss Function: $L_i((x_i, y_i), \{w, w_i\}) = \left(y_i - g(x \mid w, w_0)\right)^2$ **"Squared Loss"**

- Learning Objective: $\underset{w, w_0}{\operatorname{argmin}} \sum_{i=1}^{N} L_i((x_i, y_i), \{w, w_0\})$ Optimization Problem

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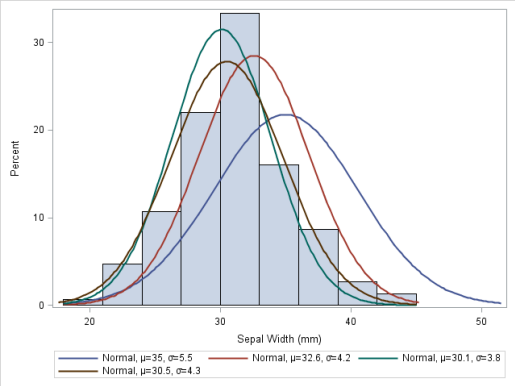

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  $L_i(x_i, \{\mu, \sigma\}) \quad = \quad ???$ **what loss to use?**

- Learning Objective: $\underset{w, w_0}{\operatorname{argmin}} \sum_{i=1}^{N} L_i((x_i, y_i), \phi)$ Optimization Problem

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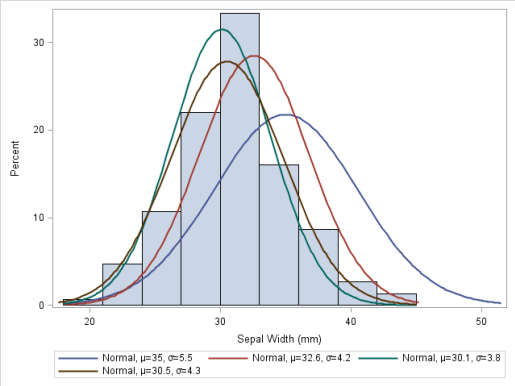
ML: main abstract ideas



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  $L_i(x_i, \{\mu, \sigma\}) = \log p(x_i; \mu, \sigma) = \log N(x_i \mid \mu, \sigma)$ **Log Likelihood loss**

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Maximum Likelihood Estimation (MLE)

This principle gives a useful, principled and widely-used loss function to estimate parameters of statistical models (from linear regression, to neural networks, and beyond).

RVs!

- Training data set: $D = \{(x_i, y_i)\}_{i=1}^{N}$ $\quad x_i \in R^D$, $y_i \in R$ or $y_i \in \{-1,1\}$

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Aside: ChatGPT uses MLE

Likelihood of all the documents on the internet:

$$\ln p(\text{Internet} \mid w) = \sum_{\text{Doc} \in \text{Internet}} \sum_{\text{Word} \in \text{Doc}} \ln p(\text{Word} \mid \text{Words before it in Doc}, w)$$

"Pre-Training": billions of $$ spent annually computing the MLE:

$$\arg\max_{w} \ln p(\text{Interent} \mid w)$$

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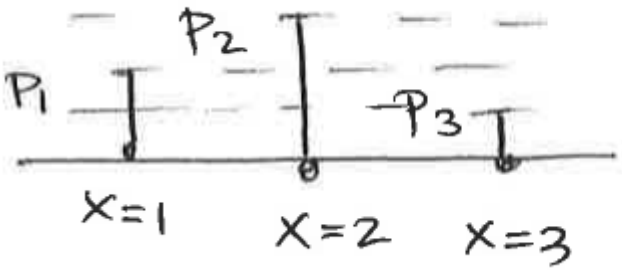
Reminder: probability distributions

Random variable (RV) distribution is a function: $\mathbf{x} \rightarrow \mathbb{R}$
e.g., $\boldsymbol{p}(x\text{=heads}) = \mathbf{0.5}$

1. Discrete RV, *e.g.*, coin toss heads/tails.
2. Continuous RV, *e.g.*, height

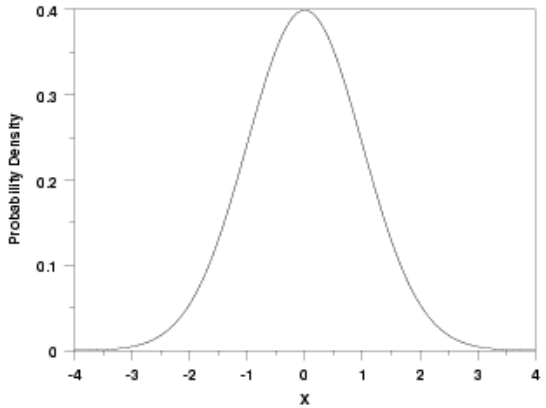
Discrete RVs have a Probability Mass Function (PMF)



sums to 1

$$P_1 + P_2 + P_3 = 1$$

Continuous RVs have a Probability Density Function (PDF)



integrates to 1

$$\int_{-\infty}^{\infty} p(x)\mathrm{d}x = 1$$

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Examples: distributions of discrete RVs

1. <u>Bernoulli RV</u>—model *one* toss of a coin that can be biased
   $P(heads) = p, \quad P(tails) = 1 - p$, parameter is p.

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Examples: distributions of discrete RVs

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2. <u>Binomial RV</u>—model *n coin tosses*, number of heads, $\mathbf{k}$

$$P(x=k) = \binom{n}{k} p^k (1-p)^{n-k}$$

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3. <u>Poisson RV</u>– model number of mutations, $\boldsymbol{k}$, occurring in a cell population with mean mutation rate, $\boldsymbol{\lambda}$, over fixed time interval

$$P(x=k) = \frac{e^{-\lambda} \lambda^k}{k!}$$

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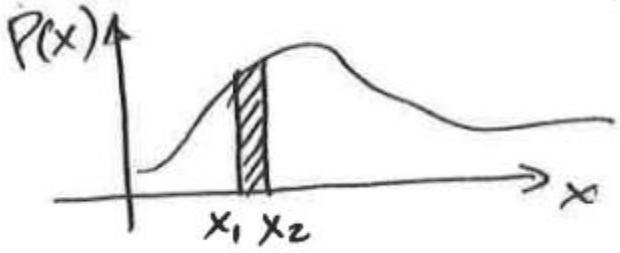
Distributions of continuous RVs

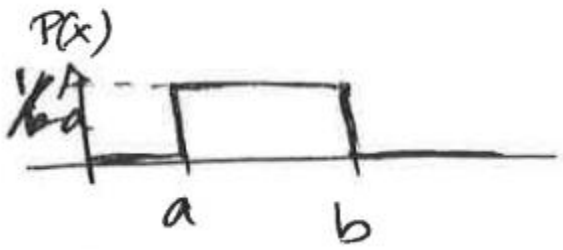
Continuous RVs have a Probability Density Function



$$\int_{x_1}^{x_2} P(x)\,dx$$ = Prob that the random variable X has a value in the interval $[x_1 \ldots x_2]$

Examples:

1. Uniform -



Parameters: a, b

2. Gaussian

$$P(x) = \frac{1}{\sqrt{2\pi}\,\sigma} \exp\left(-\frac{1}{2}\frac{(x-\mu)^2}{\sigma^2}\right)$$

Parameters: μ, σ

$$X \sim N(\mu, \sigma^2)$$

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Multivariate distributions

Space of outcomes is a vector instead of a scalar:

<u>Multinomial</u> (generalization from binomial):

- urn with balls of different colors.
- Pick a ball at random.
- p_1 it is green, p_2 it is blue and p_3 it is red

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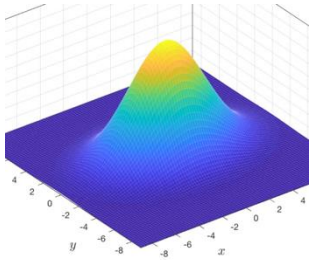
<u>Multivariate Gaussian:</u>

- Mean is a vector, and variance becomes covariance.



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The basic set-up of MLE

- Given data $D = \{x_i\}_{i=1}^{N}$ for $x_i \in R^d$
- Assume a set (family) of distributions on R^d, $\{p_\theta(x) | \theta \in \Theta\}$.

Same as $p(x|\theta)$

e.g mean (μ) and variance (σ^2) for $x \in \mathbb{R}^1$

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- Assume D contains samples from one of these distributions:

$$x_i \sim p_{\hat{\theta}}(x)$$

- This assumes that each element of D is *identically and independently distributed* (iid).

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"likelihood function"

Goal of MLE: "learn"/estimate the value of $\boldsymbol{\theta}$ that "pins down" the distribution from which the data came.

Definition: $\boldsymbol{\theta}_{MLE}$ is a MLE for $\boldsymbol{\theta}$ with respect to the data and family of distributions, if $\boldsymbol{\theta}_{MLE} = \underset{\theta \in \Theta}{\mathbf{argmax}}\, p(D|\theta)$.

function of θ.

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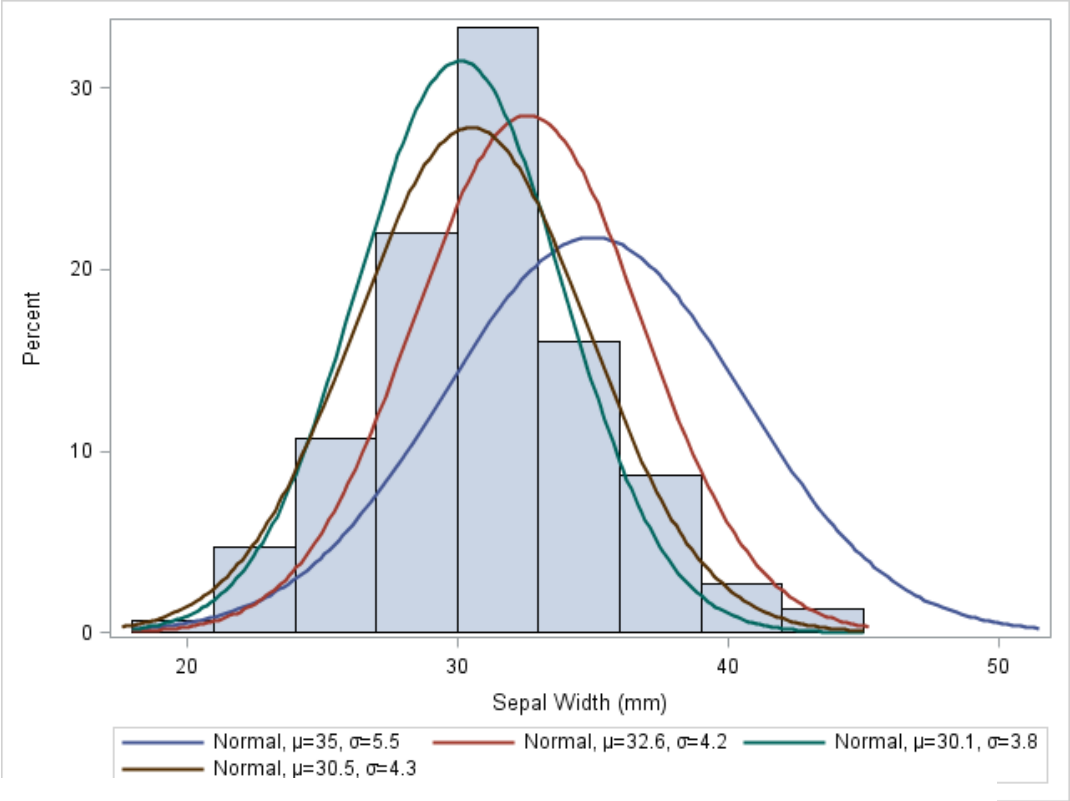
The basic set-up of MLE

$$\boxed{\theta_{MLE} = \underset{\theta \in \Theta}{\operatorname{argmax}}\, p(D|\theta)}$$



$$D = \{x_i\}_{i=1}^{N} = \{20.1, 33.8, 34.6, 36.2, \dots\}$$

Note that $p(D|\theta) = p\left(\{x_i\}_{i=1}^{N} \middle| \theta\right) = \prod_{i=1}^{N} p(x_i|\theta)$

because iid

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- Given data $D = \{x_i\}_{i=1}^{N}$ for $x_i \in R^d$
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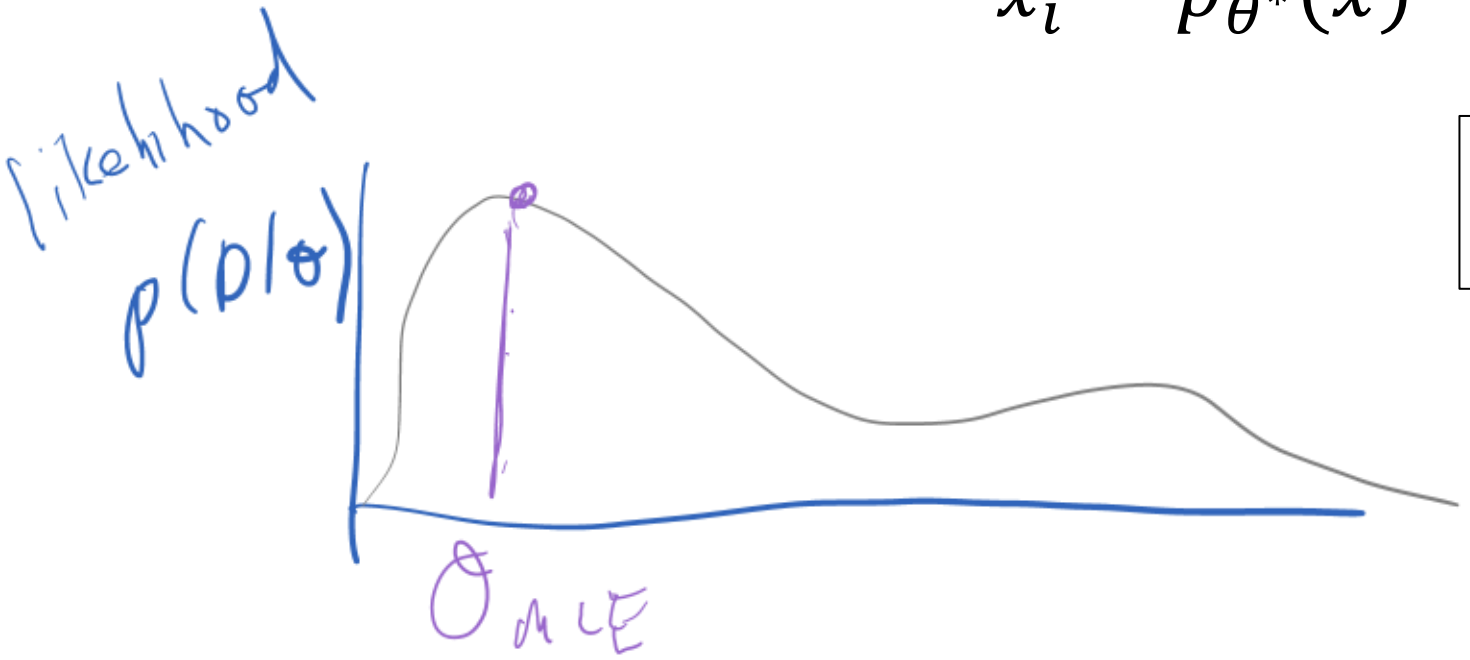
- Assume D contains samples from one of these distributions:

$$x_i \sim p_{\theta^*}(x)$$



$$\boxed{\theta_{MLE} = \underset{\theta \in \Theta}{\operatorname{argmax}}\, p(D|\theta)}$$

Is there always one unique MLE parameter value?

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Some properties of MLE

eg $N(x|\mu,\sigma)$
vs
$N(x|2+\mu,\sigma)$

- *Consistency*: as we get more and more data (drawn from one distribution in our family), then we converge to estimating the true value of $\boldsymbol{\theta}$ for $\boldsymbol{D}$.
- *Statistically efficient*: making good use of the data ("least variance" parameter estimates).
- The value of $\boldsymbol{p}(\boldsymbol{D}|\boldsymbol{\theta}_{\boldsymbol{MLE}})$ is invariant to re-parameterization.

- MLE can still yield a parameter estimate even when the data were not generated from that family (phew & caveat emptor).

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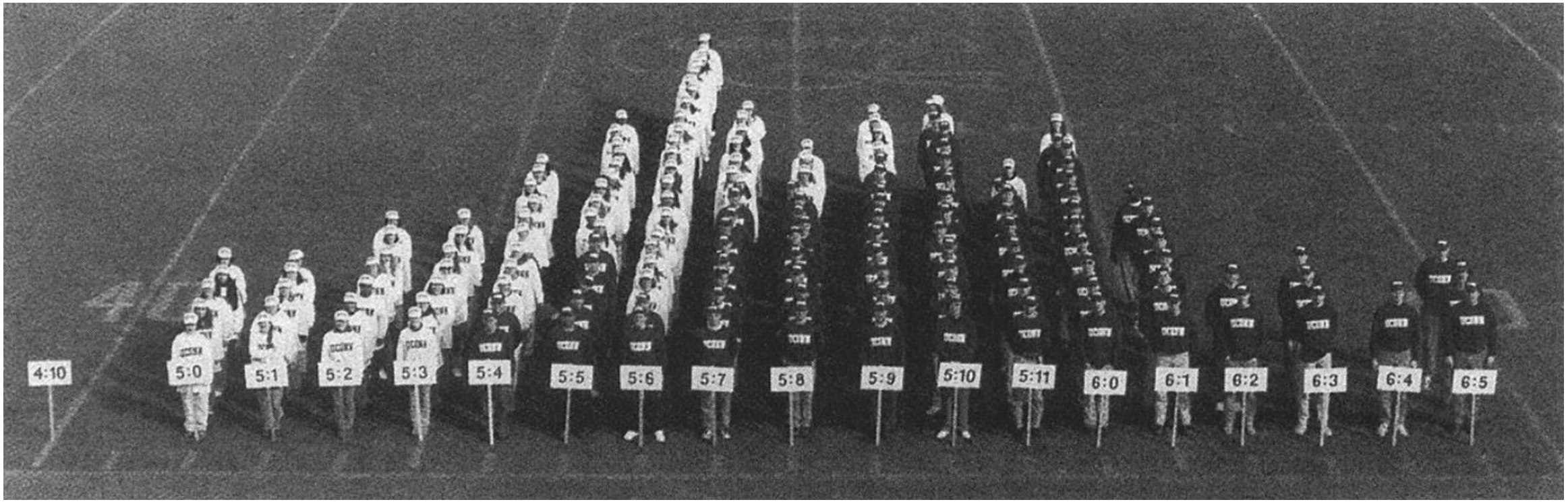
e.g. MLE for univariate Gaussian



- Arguments can be made from the Central Limit Theorem that height is normally distributed.
- Suppose you were given a set if height measurements, $\{x_i\}$, how would you derive the estimate for the mean and variance, using MLE?

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e.g. MLE for univariate Gaussian

Goal: $\theta_{MLE} = \underset{\theta \in \Theta}{\operatorname{argmax}}\, p(D|\theta)$ from set of data $D = \{x_i\}_{i=1}^{N}$

- Assume data are generated as $X \sim N(x|\mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} exp - \frac{(x-\mu)^2}{2\sigma^2}$
- So assume MLE family of distributions, $p(X = x|\theta) = N(X|\mu, \sigma^2)$.
- Now our goal is to find $\theta_{MLE} = (\mu_{MLE}, \sigma^2_{MLE}) = \underset{\theta \in \Theta}{\operatorname{argmax}}\, p(D|\mu, \sigma^2)$.
- First step, write down the likelihood function:
  - $p(D|\theta) = p(x_1, x_2, \ldots x_N|\mu, \sigma^2) = \prod_{i=1}^{N} p(x_i|\mu, \sigma^2)$.
- The product of the terms is a little inconvenient to work with, so we will take the log to get a sum.

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e.g. MLE for univariate Gaussian

- Likelihood: $p(x_1, x_2, \dots x_N | \mu, \sigma^2) = \prod_{i=1}^{N} p(x_i | \mu, \sigma^2)$.

- *Log likelihood ("LL")* is a monotonically increasing function of the likelihood.

$$\log p(D|\theta) = \sum_{i=1}^{N} \log p(x_i | \mu, \sigma^2)$$

- Therefore $\theta_{MLE} = \underset{\theta \in \Theta}{\operatorname{argmax}}\, p(D|\theta) = \underset{\theta \in \Theta}{\operatorname{argmax}}\, \log p(D|\theta)$

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e.g. MLE for univariate Gaussian

- Now we have a concrete optimization problem to work with:

$$\mu_{MLE}, \sigma^2_{MLE} = \underset{\theta \in \Theta}{\operatorname{argmax}} \log p(D|\theta) = \underset{\mu,\sigma^2}{\operatorname{argmax}} \sum_{i=1}^{N} \log p(x_i|\mu, \sigma^2)$$

- How will we solve this optimization problem?
- Find a setting of the parameters for which the partial derivatives are 0 (*i.e.*, a stationary point).
- Then check whether the setting is a maximum (negative second derivative), a minimum, etc. (first year calculus).
- (if #params>1, check if Hessian is negative definite; for 1D Gaussian Hessian is diagonal, so can check each is negative).

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e.g. MLE for univariate Gaussian

- Find the setting of the parameters that set the partial derivatives to zero:

$$\mu_{MLE}, \sigma^2_{MLE} = \underset{\theta \in \Theta}{\operatorname{argmax}} \log p(D|\theta) = \underset{\mu,\sigma^2}{\operatorname{argmax}} \sum_{i=1}^{N} \log p(x_i|\mu, \sigma^2)$$

- Lets expand out so we can take the derivative:

$$\frac{\partial}{\partial \mu} \sum_{i=1}^{N} \log p(x_i|\mu,\sigma^2) = \sum_i \frac{\partial}{\partial \mu} \log \left[\frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2\sigma^2}(x_i-\mu)^2\right)\right]$$

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$$= \sum_i \frac{\partial}{\partial \mu}\left[-\log(\sqrt{2\pi\sigma^2}) - \frac{1}{2\sigma^2}(x_i-\mu)^2\right]$$

$$= \sum_i \left[0 + \frac{1}{\sigma^2}(x_i-\mu)\right] \xRightarrow{\text{set to zero}} \sum_i x_i = \sum_i \mu$$

$$\sum_i x_i = N\mu$$

$$\Rightarrow \mu \quad \frac{\sum_i x_i}{N}$$

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e.g. MLE for univariate Gaussian

$\frac{\partial^2 (LL)}{\partial \mu^2} =$

- Find the setting of the parameters that set the partial derivatives to zero:

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set to zero

$\sum_i x_i = N\mu$

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e.g. MLE for univariate Gaussian

$$\frac{\partial^2 (LL)}{\partial \mu^2} = \sum_i \frac{1}{\sigma^2} \cdot (-1) = -\frac{N}{\sigma^2} < 0$$

$\Rightarrow$ max

$$\frac{\partial}{\partial \mu} \sum_{i=1}^{N} \log p(x_i \mid \mu, \sigma^2) = \sum_i \frac{\partial}{\partial \mu} \log \left[\frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2\sigma^2} (x_i - \mu)^2 \right) \right]$$

MLE

$$= \sum_i \frac{\partial}{\partial \mu} \left[-\log(\sqrt{2\pi\sigma^2}) - \frac{1}{2\sigma^2} (x_i - \mu)^2 \right]$$

$$= \sum_i \left[0 + \frac{1}{\sigma^2} (x_i - \mu) \right] \Rightarrow \sum_i x_i = \sum_i \mu \Rightarrow \Rightarrow \mu \quad \frac{\sum_i x_i}{N}$$

set to zero

$$\sum_i x_i = N\mu$$

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e.g. MLE for univariate Gaussian

$$N(x|\mu,\sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} exp - \frac{(x-\mu)^2}{2\sigma^2}$$

$$\mu_{MLE}, \sigma^2_{MLE} = argmax \sum_{i=1}^{N} \log N(x_i|\mu,\sigma^2)$$

- Again, but this time for σ^2:

$$\frac{\partial \mathcal{LL}}{\partial \sigma^2} = \frac{\partial}{\partial \sigma^2}\Big(-\frac{N}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum_{n=1}^{N}(x_n-\mu)^2\Big)$$

$$= -\frac{N}{2}\cdot\frac{\partial}{\partial \sigma^2}\big(\log(2\pi\sigma^2)\big) + \frac{\partial}{\partial \sigma^2}\Big(-\frac{1}{2\sigma^2}\sum_{n=1}^{N}(x_n-\mu)^2\Big)$$

$$= -\frac{N}{2}\cdot\frac{1}{2\pi\sigma^2}\cdot 2\pi + \frac{\partial}{\partial \sigma^2}\Big(-\frac{1}{2\sigma^2}\sum_{n=1}^{N}(x_n-\mu)^2\Big)$$

$$= -\frac{N}{2\sigma^2} + \sum_{n=1}^{N}\Big(-\frac{1}{2}\cdot -1\cdot(\sigma^2)^{-2}\cdot 1\cdot(x_n-\mu)^2\Big)$$

$$= -\frac{N}{2\sigma^2} + \sum_{n=1}^{N}\Big(\frac{1}{2\sigma^4}\cdot(x_n-\mu)^2\Big)$$

$$0 = \frac{1}{2\sigma^2}\Big(-N + \frac{1}{\sigma^2}\sum_{n=1}^{N}(x_n-\mu)^2\Big)$$

$$0 = -N + \frac{1}{\sigma^2}\sum_{n=1}^{N}(x_n-\mu)^2$$

$$\sigma^2 = \frac{1}{N}\sum^{N}(x_n-\mu)^2$$

σ^2_{MLE}

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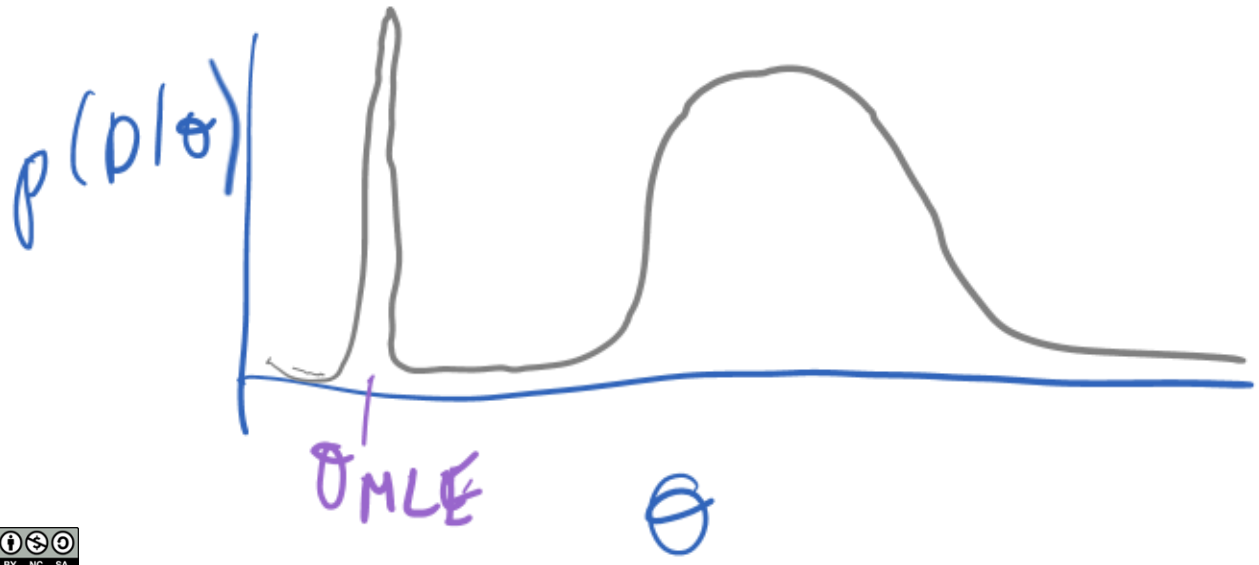
MLE yields a "point estimate" of our parameter

- When we perform MLE, we get just one single estimate of the parameter, $\boldsymbol{\theta}$, rather than a distribution over it (which captures uncertainty).
- In Bayesian statistics, we obtain a (*posterior*) distribution over $\boldsymbol{\theta}$, that is, $\boldsymbol{p(\theta|D)}$, instead of $\boldsymbol{\theta_{MLE}}$.



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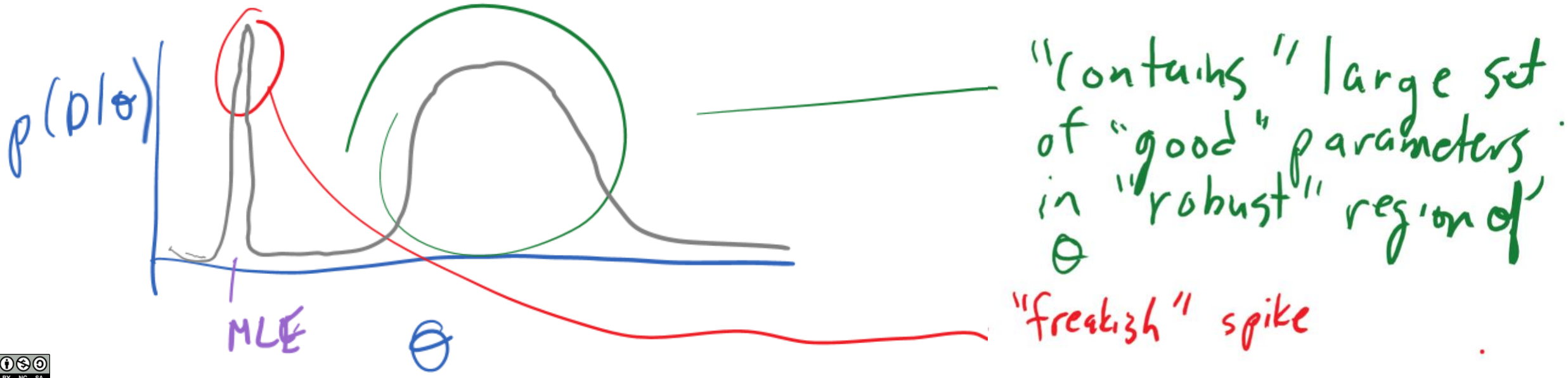

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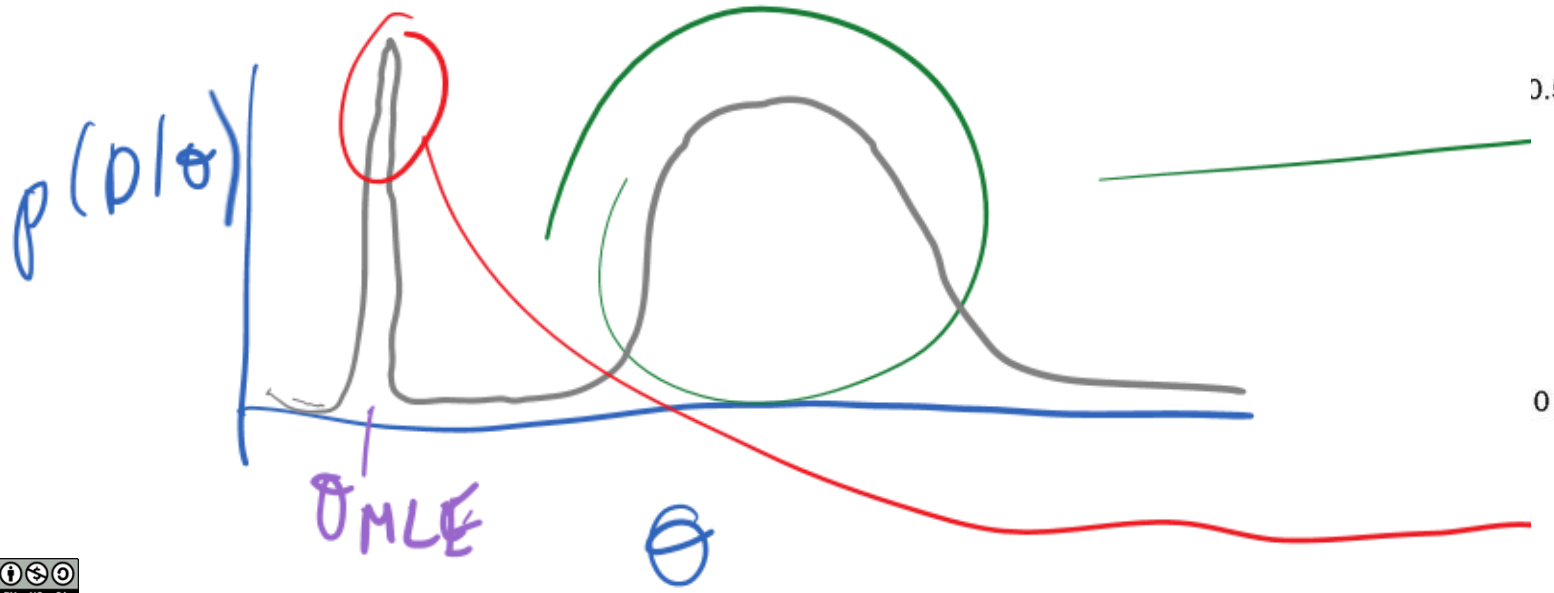

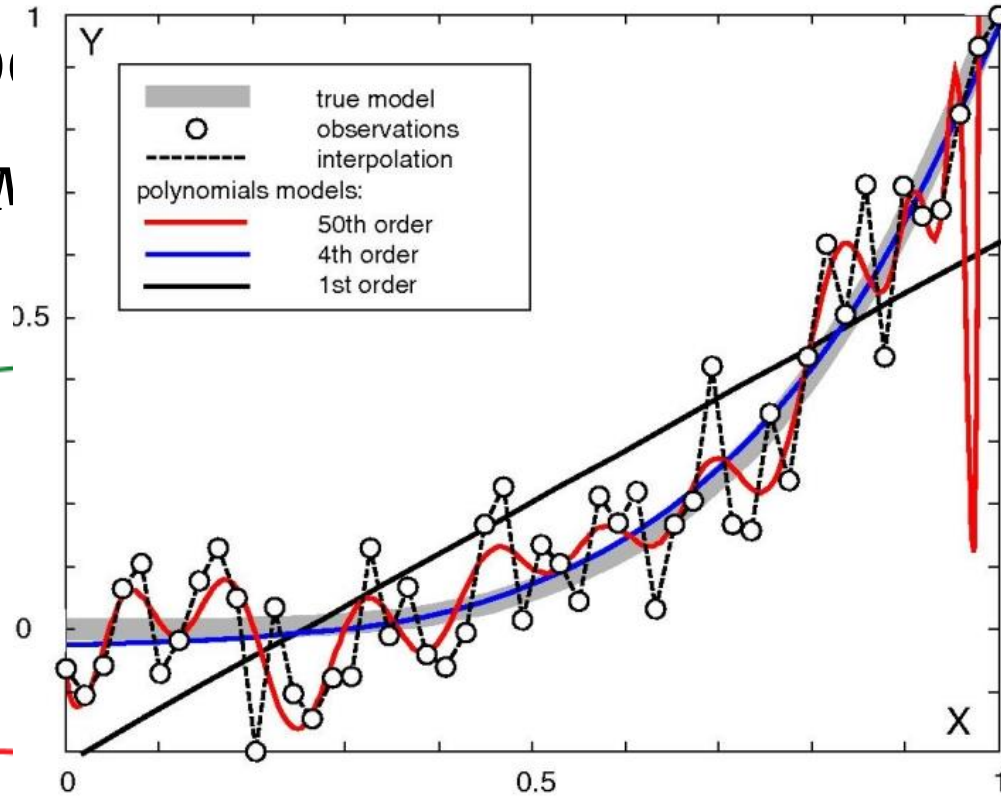

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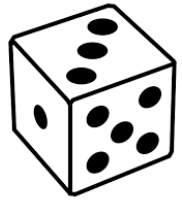
e.g. MLE for the multinomial distribution



- Consider a six-sided die that we will roll: we want to know the probability of each side of the die turning up ($\theta = \theta_1 \dots \theta_6$).
- Assume we have observed N rolls, with RV, $X \sim p_\theta(X)$.
- We write that $P(X = k|\theta) = \theta_k$ (when k^{th} side faced up).
- Lets use MLE to estimate these parameters.
- First, since one side must always face up, we know that $1 = \sum_k \theta_k$. $n_k \equiv |\{i \mid x_i = k\}|$
- Now we write the likelihood:

$$P(D|\theta) = p(x_1, \dots x_N|\theta) \quad = \prod_{i=1}^{N} p(x_i|\theta) = \prod_{i=1}^{N}\prod_{k=1}^{6} \theta_k^{I[x_i=k]} = \prod_{k=1}^{6} \theta_k^{\sum_i^N I[x_i=k]} = \prod_{k=1}^{6} \theta_k^{n_k}$$

Now our MLE problem becomes:

$$\theta_{MLE} = \underset{\theta\in\Theta}{\operatorname{argmax}} \log p(D|\theta) = \underset{\theta\in\{\Theta|1=\sum_k \theta_k\}}{\operatorname{argmax}} \sum_{k=1}^{6} \log \theta_k^{n_k}$$

constrained optimization

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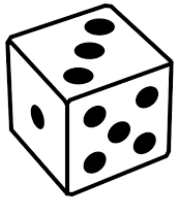
e.g. MLE for the multinomial distribution

$n_k = |\{i \mid x_i = k\}|$



Have a constrained optimization problem:

constrained optimization

$$\theta_{MLE} = \underset{\theta \in \Theta}{\operatorname{argmax}} \log p(D|\theta) = \underset{\theta \in \{\Theta | 1 = \sum_k \theta_k\}}{\operatorname{argmax}} \sum_{k=1}^{6} \log \theta_k^{n_k}$$

What is one technique you should have learned in first year calculus to solve this?

The technique of Lagrange multipliers (Appendix C of textbook):

$J(\theta, \lambda) = \log p(D|\theta) + \lambda(1 - \sum_k \theta_k)$ (look for stationary points wrt θ, λ)

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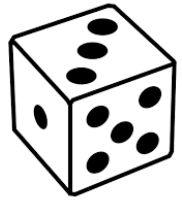
e.g. MLE for the multinomial distribution



$n_k = |\{i \mid x_i = k\}|$

$$J(\theta, \lambda) = log\ p(D|\theta) + \lambda(1 - \sum_k \theta_k) = \sum_{k=1}^{6} \log \theta_k^{n_k} + \lambda(1 - \sum_k \theta_k)$$

1. $\frac{\partial J}{\partial \lambda} = 0 \Rrightarrow 1 = \sum_k \theta_k$ (we just get the constraint back)

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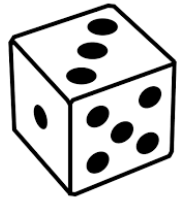
e.g. MLE for the multinomial distribution



$$J(\theta,\lambda) = log\ p(D|\theta) + \lambda(1 - \sum_k \theta_k) = \sum_{k=1}^{6} \log \theta_k^{n_k} + \lambda(1 - \sum_k \theta_k)$$

1. $\frac{\partial J}{\partial \lambda} = 0 \Rightarrow 1 = \sum_k \theta_k$ (we just get the constraint back)
2. $\frac{\partial J}{\partial \theta_j} = \frac{\partial}{\partial \theta_j} \sum_{k=1}^{6} \log \theta_k^{n_k} - \frac{\partial}{\partial \theta_j} \lambda \theta_j$

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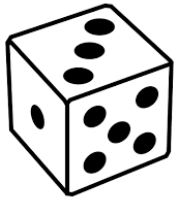
e.g. MLE for the multinomial distribution



$$J(\theta,\lambda) = log\ p(D|\theta) + \lambda(1 - \sum_k \theta_k) = \sum_{k=1}^{6} \log \theta_k^{n_k} + \lambda(1 - \sum_k \theta_k)$$

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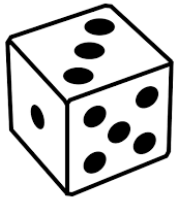
e.g. MLE for the multinomial distribution



$$J(\theta,\lambda) = log\ p(D|\theta) + \lambda(1 - \sum_k \theta_k) = \sum_{k=1}^{6} \log \theta_k^{n_k} + \lambda(1 - \sum_k \theta_k)$$

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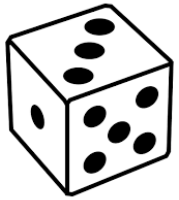
e.g. MLE for the multinomial distribution



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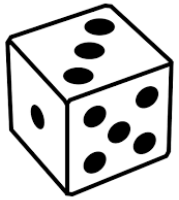
e.g. MLE for the multinomial distribution



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3. Lets plug this into 1), $1 = \sum_k \theta_k = \sum_k \frac{n_k}{\lambda} \quad \Rrightarrow \quad \lambda = \sum_k n_k = N.$

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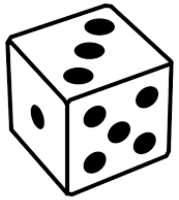
e.g. MLE for the multinomial distribution



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3. Lets plug this into 1), $1 = \sum_k \theta_k = \sum_k \frac{n_k}{\lambda} \quad \Rrightarrow \quad \lambda = \sum_k n_k = N.$
4. All together then, $\theta_k = \frac{n_k}{N}.$

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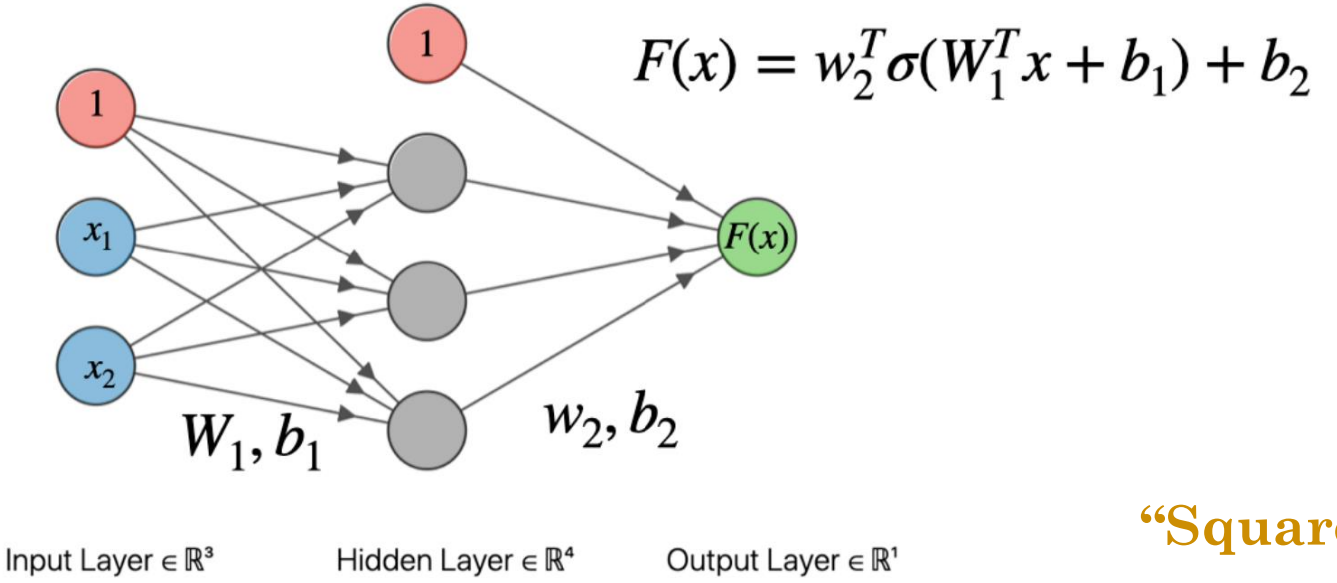
Doing MLE requires optimization

$$\theta_{MLE} = \underset{\theta \in \Theta}{\operatorname{argmax}} \log p(D|\theta)$$

- For Gaussians, multinomials, & linear regression, the MLE can be obtained *in closed form* by setting the derivative to zero: *e.g.*, $\theta_k = \frac{n_k}{N}$.
- What if we had a neural network model?



Here, we will need *iterative optimization* (gradient descent).

"Squared Loss" derived from MLE

$$L_i((x_i, y_i), \{W_1, w_2, b_1, b_2\}) = \left(y_i - g(x \mid W_1, w_2, b_1, b_2)\right)^2$$

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